Developing synthetic data for FIWARE-enabled Digital Twins in the water sector

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ABSTRACT

Digital twins for the water sector are opening new methods of research for water engineers. Languages like Python with its rich set of plug-in package functionality make it relatively straightforward for researchers to take raw data and process it using a rich variety of off-the-shelf data classification, evaluation, and clustering models to create information and knowledge from data, before presenting it using a variety of presentation frameworks.

This approach creates two clear challenges: 1) proving that complex data processing pipelines work as intended and 2) collecting the 'right' kind of source data, particularly for edge cases and leaks.

In this work, we are reviewing our experiences from aqua3S, Fiware4Water and LOTUS to show how we have developed and used synthetic data approaches to 1) demonstrate data integrity across FIWARE-based data processing pipelines, 2) demonstrate how volume data can be processed within real world time constraints and 3) create suitable edge-case data for testing the operation of algorithms.

Through our work on these projects, we have started to develop a wealth of best-practice, strategic choices, and approaches to be avoided for successful and long-term context broker-based data management.

Keywords: fiware, digital twins, smart water, synthetic data, data science

1 Introduction

The purpose of this paper is to capture a set of good practice approaches for working with synthetic data generation in digital twin projects, drawn from our involvements with the aqua3S [1] , Fiware4Water [2] and LOTUS [3] projects. It is expected that these practices can then be applied to our current Waterverse [4] project.

We can define a 'digital twin' as a virtual representation of a real-world model. From our experiences in the water sector, this can typically be a water distribution network, such as those modelled by EPANET [5] or EPA SWMM [6], or numerical/statistical models developed from previously observed physical model operations.

The real value of the digital twin is that the accurate virtualised model enables researchers to explore scenarios which may occur infrequently in a real-world model, e.g., leaks, high consumption, floods and so on. Typically, this data may be used as training and testing data for developing machine learning models [7].

The challenges of working with digital twins can be broken into two key areas: data management and data processing, in that a digital twin may generate a lot of data that needs to be stored and, regardless of the amount of data generated by the twin, it will need to be processed.

For managing data on these projects, we have used FIWARE [8], both the Stellio and Orion variants.

For processing data, we have considered three key use-cases: 1)proof-of-concept, data can be collected, stored in FIWARE and processed in a suitable order, 2) scale-of-process, algorithms developed to process data are able to complete processing within the time and processing constraints for the results generated to have value, 3) edge-casing, in that suitable data can be created to simulate desired atypical or extraordinary conditions.

2 Materials and Methods

In this section of the paper, we will outline the functionality of the FIWARE broker and the operation of the existing projects (aqua3S, Fiware4Water and LOTUS) and their requirements for synthetic data generation.

2.1 FIWARE broker

FIWARE brokers are implementations of CMI-based standards [9] for ngsi-ld/v1 or ngsi-ld/v2, describing a CRUD-based data lifecycle of create, retrieve, update, and delete for smart models, realised as JSON-LD key-value pairs. FIWARE maintains a repository of smart models [10], that developers can use, extend or re-implement for their projects, The broker interface is realised through HTTP requests using post-based requests for smart model creation, get for retrieval, patch to update, and delete for deletion.

For the projects described in this paper, broker functionality was implemented through Stellio and Orion-LD / Cygnus and although the brokers were provided by different providers, their functional interfaces (through nsgi-ld/v1) were largely identical.

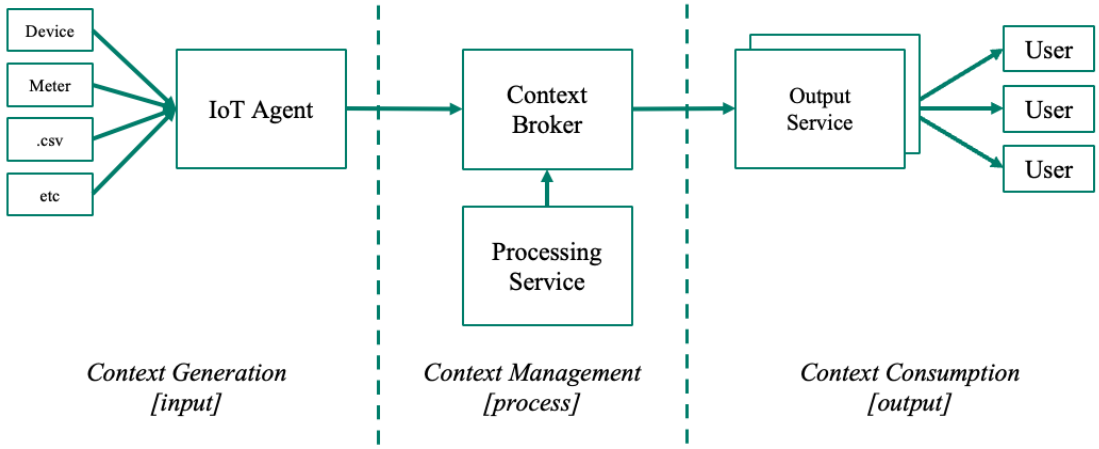

Figure 1 Prototypical FIWARE architecture

Figure 1 outlines a prototypical FIWARE implementation with three core functions: context generation, context management and context consumption. For digital twins, real-world data is collected as inputs to the IoT Agent in the context generation phase. The goal of this phase is to collect data and create / generate smart model instance (contexts) representations of the data to be stored in the broker. For synthetic data generation activities, it is this functionality that is replaced such that the broker no longer collects real-world data but, instead, collects generated data.

The context management function of the broker describes functionality where contexts can be processed to create new data and/or contexts. Typically, this describes activities where 'dumb' data is collected from the IoT Agent and then processed, potentially with other data, to create insights, e.g., sensor data may be collected by the IoT Agent and is processed later to determine is the values received are in or out of range.

Finally, the context consumption function describes activities where broker data is processed as output. This is likely to be situation specific and may include activities like producing daily reports, providing data for webapps and so on.

2.2 Aqua3S

The aqua3S project was concerned with the standardisation, security and safety of water networks and achieved this through the development of digital twins for 8 pilots. Data was collected across a range of real-world sensors and stored in a broker and processed to determine if sensor readings were 'out of range' and raise alarms. The system provided multiple visualisations source and processed data in geographic, chart-based, and textual forms, Figure 2.

Figure 2 aqua3S visualisations

To develop the visualisation, early warning, and decision support components for the project, it was essentially to initially consider a proof of concept for the data pipeline provided by FIWARE, that was to ensure that the data that was coming out of the broker as context consumption was the same as the data going into it. To realise this, labelled synthetic data was created, injected into the broker, and then collected, initially at *context consumption* and then at *context management* stages.

Scale of process became an issue with the development of historical data processing, in that charting functionality was required to provide up to 1 year of historic data. This data was generated synthetically as the project had not been running long enough to collect data organically. Attempting to chart this long-term data *on-demand* demonstrated that the naïve algorithm was unsuitable and was redeveloped into a backend process.

To develop alert, anomaly and leak detection functionality, the project needed to collect out of range and leaky data. Unfortunately, the pilots were all far too reliable to provide this type of data organically, with one pilot operating stating that they tended to have one alert per year.

EPANET data was used as a source for leak detection and localisation algorithm development, given that pilot water distribution networks generally had low leak incident rates. This enabled us to develop a hybrid approach where leak algorithms could be trained off synthetic data and then used on real-world data. However, EPANET was limited to flow and pressure attributes and could not be used across all sensors.

For other sensors, the existing anomaly detection system was used. For this system, historic sensor readings were collected and bucketed into 8-hour slots to determine average and min/max values. For normal anomaly detection, novel readings were compared against these min/max values and an anomaly was raised. Likewise, if readings were outside of the operator-provided alert ranges, an alert was raised, Figure 3.

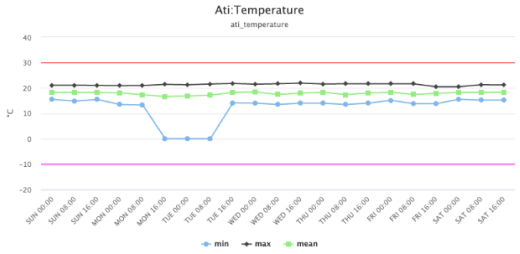

Figure 3 aqua3S sensor anomaly record

This data was co-opted for synthetic data generation, in that for 'normal' data, values could be calculated inside of the anomaly min/max range, for anomaly generating data, the values would be outside of the anomaly range but inside the alert range and for alerts, the values would be outside of the alert range. This approach was particularly useful for generating plausible 'in-range' data.

2.3 Fiware4Water

For the Fiware4Water project, University of Exeter partnered with South West Water (SWW), a water supply company in the UK, for a smart metering pilot where consumption data was collected via FIWARE. Data was processed and provided to customers through a smart app, and to SWW for leak detection interventions through continuous flow alarms and atypical consumption patterns.

Prior to our involvement on the project, a Stellio broker had been installed and was already collecting meter data. Although having a working data pipeline in place was a huge benefit for the project, it did create some issues, namely: for us, it was our first involvement with FIWARE, and we needed to be able to develop a proof of concept to understand how Stellio and FIWARE worked. In addition, the broker had only recently been installed and had limited historic data and was a 'live' system. Of course, we would rather do development and testing work in separate environments as to not risk the live data.

To develop our understanding of Stellio and as a proof of concept, we configured a test Stellio container and filled it with labelled data which was then accessed to demonstrate stability / robustness of context generation, processing, and consumption.

Like the aqua3S project, this project had a 12-month event horizon for historic data. Having separate development and test broker allowed for us to create 'scale of process' data, which was particularly beneficial for the social functionality of the smart app, where customer consumption was ranked against neighbourhood consumption. This algorithm required multiple broker requests across the entire population, which was initially an incredibly time-consuming operation for Stellio. Developing with scale data allowed us to refactor this into a far more efficient algorithm.

The edge cases for this project were focused on unusual consumption patterns, missing consumption readings and the occurrence of flow alarms. Although this data was generated organically, it tended to be hard to manage reliably, as live data often didn't match the patterns that were being searched for. Using synthetic edge case generation made it far easier to generate data reliably to test escalation functionality, e.g., raising alarms after N-days of no readings or flow alarms.

2.4 LOTUS

For the LOTUS project, we partnered with the Indian Institute of Technology Guwahati to develop leak detection algorithms using their local water distribution network. The LOTUS project is somewhat different to the previous two projects in that it is on-going and that it relies on synthetic data as there is no organic / live aspect to the project currently, though this will be addressed later in the project.

To develop synthetic data for this project, EPANET was employed a digital twin. A cross-platform (Windows & Linux) desktop digital twin was developed that enabled the water network to be run on-demand and develop leaks on an ad hoc basis. This approach enabled both research groups to develop leak detection approaches in isolation and compare results.

Like the other two projects, initial proof of concept data pipeline work was undertaken to ensure that the data coming out of the broker was the same as the data going in. This was complicated by the additional need to demonstrate that the results being generated by the EPANET Python library matched results being generated by the EPANET application.

Developing synthetic data at scale produced some interesting results. Initially, we attempted to store the entire Guwahati EPANET model in the broker, some 300 entities (links and nodes) with 8 and 4 attributes respectively. However, this provided to be unworkable given the number of HTTP requests to patch data, particularly when multiple simulation steps were performed. Instead, we moved to a simpler model where only simulated sensors were stored rather than the entire

network and only required attributes (flow and pressure). This gave far better performance and more closely represented real-world situations where water distribution networks have a small number of sensors. Having a relatively short turnaround for network simulation meant that it became trivial to build alternate sensor configurations and populate FIWARE with test data.

The key edge case for this project was the injection of leaks into the network. This was realised using the EPANET emitter equation and was built into a user testbed which enabled users to start and stop leaks on demand and then simulate the water network in a stepwise fashion to observe the leak as its impact progressed through the network data.

3 Results

3.1 Proof of concept data generation

For proof-of-concept, it was vital to demonstrate that the data that came out of FIWARE was as expected. Whilst this was clear for aqua3S and LOTUS, where the data pipeline was built from scratch, it became apparent for the Fiware4Water project that had an existing live dataset. In this instance, it was easy to fall into the trap of assuming that data was correct as each data request would generally return data, though not necessarily desired data. To address all the projects, labelled synthetic data was created that could easily be followed through our processes.

3.2 Scale of process data generation

For scale-of-process, there were several cases where the time required to process data grew as data loads increased to the point where the time taken was unacceptable, particularly in HTTP request processing. To address this, algorithms were developed that could generate bulk data to simulate normal operation, particularly for generating the year range of sensor chart data in aqua3S. In most cases algorithms were fundamentally re-developed, typically moving to threaded and/or cached solutions.

3.3 Edge case data generation

For edge-casing, there were many situations where *out of range* data was required, if for nothing more than to test algorithms in development. Whilst arbitrary out of range data could be injected into a pilot, often data quantity and quality were closely related. For aqua3S, it was discovered that historic physical sensor data could be re-used to create alert and anomaly edge case data, demonstrating that not all synthetic data needs to be purely model based. EPANET was typically used for data generation and leak injection. However, this did limit data to flow and pressure attributes.

4 Conclusion and Perspectives

From the three projects presented, there is clear value in the creation and use of synthetic data generation across the three modalities discussed and a synthetic data generation tool for the Waterverse project should look to capture this functionality into a package that can be re-used across its pilots. The pilots work across water distribution and storm and wastewater management, creating a create need to integrate EPANET and EPA SWMM into synthetic data generation functionality, particularly regarding the generation of data at scale and edge cases. However, some pilots are likely to require additional quantitative models, so the provision of EPANET and EPA SWMM shouldn't be viewed as a closed solution.

4.1 FIWARE integration

Regarding the FIWARE model, Figure 1, it would appear logical to replace an existing IoT Agent with the synthetic data generation tool as this provides a functional equivalence to normal / organic operation and it should be possible to replace the synthetic tool with the organic IoT Agent on-demand. It is highly likely that the synthetic tool will require a degree of configuration, e.g., to define sensors, their ranges and reporting periods, and, potentially, user control to create instance data on demand.

4.2 Delivery / Deployment of functionality

This functionality could be packaged as webserver, providing user functionality through a web interface, which is an approach that has been taken on previous projects (aqua3S). However, for more flexibility in use, it is recommended that the core synthetic data generation functionality is built into a reusable package / library (assuming Python) that can be deployed into both webapp and standalone app environments to give end users more flexibility in operations, not least as FIWARE smart models are often modified between projects and it would be highly advantageous to not inadvertently dictate model specifications.

5 Acknowledgements

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